

Série 1

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① a) 2/2

Como a relação de comutação independe de representação, demonstre para uma representação específica.

$$X^{\prime m} = X^m + W^m_{\nu} X^{\nu}$$

$$= X^m + W_{\alpha\beta} (g^{\alpha m} X^{\beta})$$

$$= X^m + \frac{W_{\alpha\beta}}{2} (g^{\alpha m} X^{\beta} - g^{\beta m} X^{\alpha})$$

$$= X^m + \frac{i}{2} W_{\alpha\beta} (-i) (g^{\alpha m} g^{\beta}_{\nu} - g^{\beta m} g^{\alpha}_{\nu}) X^{\nu}$$

$$= X^m + \frac{i}{2} W_{\alpha\beta} (\gamma^{\alpha\beta})^m_{\nu}$$

$$(\gamma^{\alpha\beta})^m_{\nu} = -i (g^{\alpha m} g^{\beta}_{\nu} - g^{\beta m} g^{\alpha}_{\nu}) \quad \text{OK}$$

$$[\gamma^{\alpha\beta}, \gamma^{\rho\sigma}] =$$

$$= - (g^{\alpha m} g^{\beta}_{\lambda} - g^{\beta m} g^{\alpha}_{\lambda}) (g^{\rho\lambda} g^{\sigma}_{\nu} - g^{\sigma\lambda} g^{\rho}_{\nu}) + (g^{\rho m} g^{\sigma}_{\lambda} - g^{\sigma m} g^{\rho}_{\lambda}) (g^{\alpha\lambda} g^{\beta}_{\nu} - g^{\beta\lambda} g^{\alpha}_{\nu})$$

$$= - g^{\alpha m} g^{\rho\beta} g^{\sigma}_{\nu} + g^{\alpha m} g^{\sigma\beta} g^{\rho}_{\nu} + g^{\rho m} g^{\rho\alpha} g^{\sigma}_{\nu} + g^{\rho m} g^{\sigma\alpha} g^{\rho}_{\nu} + g^{\rho m} g^{\alpha\sigma} g^{\beta}_{\nu} - g^{\rho m} g^{\beta\sigma} g^{\alpha}_{\nu} - g^{\sigma m} g^{\alpha\rho} g^{\beta}_{\nu} + g^{\sigma m} g^{\rho\rho} g^{\alpha}_{\nu}$$



$$= g_{\sigma\beta} (g_{\alpha\mu} g_{\rho\nu} - g_{\rho\mu} g_{\alpha\nu}) + \\ + g_{\rho\alpha} (g_{\beta\mu} g_{\sigma\nu} - g_{\sigma\mu} g_{\beta\nu}) + \\ + g_{\rho\beta} (g_{\sigma\mu} g_{\alpha\nu} - g_{\alpha\mu} g_{\sigma\nu}) + \\ + g_{\alpha\sigma} (g_{\rho\mu} g_{\beta\nu} - g_{\beta\mu} g_{\rho\nu})$$

$$i[\gamma^{\alpha\beta}, \gamma^{\rho\sigma}] = g_{\sigma\beta} \gamma^{\rho\alpha} + g_{\rho\alpha} \gamma^{\sigma\beta} + \\ + g_{\rho\beta} \gamma^{\alpha\sigma} + g_{\alpha\sigma} \gamma^{\rho\beta}$$

OK

$$\textcircled{1} b) \gamma^i \equiv \frac{1}{2} \epsilon^{ijk} \gamma^{jk}$$

$$K^i \equiv \gamma^{0i}$$

$$[\gamma^i, \gamma^j] = \frac{1}{4} \epsilon^{ikl} \epsilon^{jmn} [\gamma^{kl}, \gamma^{mn}] =$$

$$= \frac{1}{4} \epsilon^{ikl} \epsilon^{jmn} i (g^{lm} \gamma^{kn} - g^{km} \gamma^{ln} + \\ - g^{ln} \gamma^{km} + g^{kn} \gamma^{lm})$$

$$= \frac{i}{4} (-\epsilon^{ikm} \epsilon^{jmn} \gamma^{kn} + \epsilon^{jml} \epsilon^{ikn} \gamma^{ln} + \\ + \epsilon^{ikn} \epsilon^{jml} \gamma^{km} - \epsilon^{jnl} \epsilon^{ikm} \gamma^{lm})$$



$$= \frac{i}{4} \left[ -(\delta^{in} \delta^{kj} - \delta^{ij} \delta^{kn}) \gamma^{kn} + \right. \\ \left. + (\delta^{ln} \delta^{ij} - \delta^{lj} \delta^{in}) \gamma^{ln} + \right. \\ \left. + (\delta^{ij} \delta^{km} - \delta^{im} \delta^{kj}) \gamma^{km} + \right. \\ \left. - (\delta^{lj} \delta^{im} - \delta^{lm} \delta^{ij}) \gamma^{lm} \right]$$

$$= \frac{i}{4} \left[ -\gamma^{ii} + \gamma^{nn} \delta^{ij} + \gamma^{nn} \delta^{ij} + \right. \\ \left. - \gamma^{ii} + \gamma^{mm} \delta^{ij} - \gamma^{ii} - \gamma^{ii} + \right. \\ \left. + \gamma^{mm} \delta^{ij} \right]$$

→ Antisimétrico:  $\gamma^{nn} = 0$  or.

podria ter metodo que  $J^{ij} = \epsilon^{ijk} J^k$

$$[\gamma^i, \gamma^j] = +i \gamma^{ij} = +i \epsilon^{ijk} \gamma^k$$

$$[\gamma^i, \gamma^j] = \frac{1}{2} \epsilon^{ikl} [\gamma^{kl}, \gamma^{ij}] =$$

$$= \frac{1}{2} \epsilon^{ikl} i (\gamma^{kl} \gamma^{ij} - \gamma^{ij} \gamma^{kl} + \\ - \gamma^{lj} \gamma^{ki} + \gamma^{ki} \gamma^{lj})$$



$$\begin{aligned}
 [\gamma^i, K^j] &= \frac{\hbar}{2} \epsilon^{ikl} (\delta^{lj} \gamma_{k0} - \delta^{kj} \gamma_{l0}) \\
 &= \frac{\hbar}{2} (\epsilon^{ikj} \gamma_{k0} - \epsilon^{ijl} \gamma_{l0}) \\
 &= \hbar \epsilon^{ijl} K^l
 \end{aligned}$$

$$\begin{aligned}
 [K^i, K^j] &= \hbar (g^{i0} \gamma^{0j} - g^{00} \gamma^{ij} + \\
 &\quad - g^{ij} \gamma^{00} + g^{0j} \gamma^{i0})
 \end{aligned}$$

$$[K^i, K^j] = -\hbar \gamma^{ij} = -\hbar \epsilon^{ijk} \gamma_k$$

$$\gamma_{\pm}^i \equiv \frac{1}{2} (\gamma^i \pm \hbar K^i)$$

$$\begin{aligned}
 [\gamma_+^i, \gamma_+^j] &= \frac{1}{4} ([\gamma^i, \gamma^j] + \hbar [\gamma^i, K^j] + \\
 &\quad + \hbar [K^i, \gamma^j] - [K^i, K^j])
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{4} (\hbar \epsilon^{ijk} \gamma_k + \hbar \hbar \epsilon^{ijk} K^k + \\
 &\quad + \hbar (-\hbar) \epsilon^{jik} K^k - (-\hbar) \epsilon^{ijk} \gamma_k)
 \end{aligned}$$



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$$= \frac{i}{2} \epsilon^{ijk} (\gamma^k + i K^k)$$

$$[\gamma_+^i, \gamma_+^j] = i \epsilon^{ijk} \gamma_+^k$$

$$[\gamma_-^i, \gamma_-^j] = \frac{1}{4} ([\gamma_-^i, \gamma_-^j] - i [\gamma_-^i, K_-^j] + \\ - i [K_-^i, \gamma_-^j] - [K_-^i, K_-^j])$$

$$= \frac{1}{4} (i \epsilon^{ijk} \gamma_-^k - i i \epsilon^{ijk} K_-^k + \\ - i (-i) \epsilon^{jik} K_-^k - (-i) \epsilon^{ijk} \gamma_-^k)$$

$$= \frac{i}{2} \epsilon^{ijk} (\gamma_-^k - i K_-^k)$$

$$[\gamma_-^i, \gamma_-^j] = i \epsilon^{ijk} \gamma_-^k$$

$$[\gamma_+^i, \gamma_-^j] = \frac{1}{4} ([\gamma_+^i, \gamma_-^j] - i [\gamma_+^i, K_-^j] + \\ + i [K_-^i, \gamma_-^j] - i^2 [K_-^i, K_-^j])$$



$$= \frac{1}{4} (\cancel{i \varepsilon^{ijk} j^k} - \cancel{i \hat{n} \varepsilon^{ijk} K^k} + \cancel{i(-i) \varepsilon^{ijk} K^k} - \cancel{i \varepsilon^{ijk} j^k})$$

$$[j^i, j^j] = 0$$

ou  $so^+(1,3)$  é não compacto, enquanto  $su(2) \oplus su(2)$  é compacto  
 o isomorfismo correto é  $so^+(1,3) \cong su(2) \oplus su(2)$   
 → isso não é verdade

$$\underline{SO(1,3) \approx SU(2) \oplus SU(2)} \quad \text{CUIDADO!}$$

As representações irreduzíveis são:

$$j^k = j_+^k + j_-^k$$

podemos ter comentado  
 que as repr de  $su(2)$   
 são indexados por  
 semi-inteiros.

| $j_+$         | $j_-$         | $j$           |
|---------------|---------------|---------------|
| 0             | 0             | 0             |
| $\frac{1}{2}$ | 0             | $\frac{1}{2}$ |
| 0             | $\frac{1}{2}$ | $\frac{1}{2}$ |
| $\frac{1}{2}$ | $\frac{1}{2}$ | $0 \oplus 1$  |

⋮



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$$\textcircled{2} a) \quad \mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$\frac{\partial \mathcal{L}}{\partial A_\beta} - \partial_\alpha \frac{\partial \mathcal{L}}{\partial \partial_\alpha A_\beta} = 0 \quad (E-L)$$

$$\frac{\partial \mathcal{L}}{\partial A_\beta} = 0$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \partial_\alpha A_\beta} &= -\frac{1}{4} 2 F^{\mu\nu} (\delta_\mu^\alpha \delta_\nu^\beta - \delta_\nu^\alpha \delta_\mu^\beta) \\ &= -\frac{1}{2} (F^{\alpha\beta} - F^{\beta\alpha}) \\ &= -F^{\alpha\beta} \end{aligned}$$

$$\partial_\alpha F^{\alpha\beta} = 0 \quad // \quad \text{or}$$



$$\textcircled{2} b) A'_m(x^\nu + a^\nu) = A_m(x^\nu)$$

$$\delta A_m = A'_m(x^\nu) - A_m(x^\nu)$$

$$= A_m(x^\nu - a^\nu) - A_m(x^\nu) = -a^\nu \partial_\nu A_m$$

OK

$$\delta x^\mu = a^\mu$$

$$\partial_m \left( \frac{\partial \mathcal{L}}{\partial \partial_m A_\nu} \delta A_\nu + \delta x^\mu \mathcal{L} \right) = 0 \quad ??$$

$$\partial_m \left( -F^{\mu\nu} (-a^\alpha \partial_\alpha A_\nu) + a^\mu \mathcal{L} \right) = 0$$

$$\partial_m \left[ F^{\mu\nu} \partial^\alpha A_\nu + g^{\alpha m} \left( \frac{1}{4} \right) F^{\alpha\delta} F_{\delta\alpha} \right] = 0$$

$$\partial_m \left[ -F^{\mu\alpha} \partial^\nu A_\alpha + \frac{1}{4} g^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta} \right] = 0$$

$$\partial_m T^{\mu\nu} = 0$$

$$T^{\mu\nu} = -F^{\mu\alpha} \partial^\nu A_\alpha + \frac{1}{4} g^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta}$$

OK.... poderia ter calculado antes  $T^{\mu\nu}$  pois depois ter verificado a conservação.



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proq 9

$$(3) S' = \int d^4x' \mathcal{L}(\psi', \partial_\mu \psi')$$

$$d^4x' = e^{\epsilon} d^4x$$

$$\frac{\partial}{\partial x'} = \frac{\partial x}{\partial x'} \frac{\partial}{\partial x} = e^{-\epsilon} \frac{\partial}{\partial x}$$

$$S' = \int e^{\epsilon} d^4x \left[ \frac{1}{2} e^{-\epsilon} \partial_\mu (e^{-\epsilon} \psi) e^{-\epsilon} \partial^\mu (e^{-\epsilon} \psi) + \right. \\ \left. - \frac{\lambda}{4!} e^{-4\epsilon} \psi^4 \right]$$

$$S' = \int d^4x \left( \frac{1}{2} \partial_\mu \psi \partial^\mu \psi - \frac{\lambda}{4!} \psi^4 \right) = S$$

é transformação de simetria!

$$\psi'(e^\epsilon x) = e^{-\epsilon} \psi(x)$$

$$\psi'(x) = e^{-\epsilon} \psi(e^{-\epsilon} x)$$

→ infinitesimal:

$$\psi'(x) = (1 - \epsilon) \psi((1 - \epsilon)x)$$

$$= \psi(x) - \epsilon x^\mu \partial_\mu \psi(x) - \epsilon \psi(x) + O(\epsilon^2)$$



$$\delta \varphi = \varphi'(x) - \varphi(x)$$

$$= -\varepsilon [X^m \partial_m \varphi + \varphi] \quad \text{or}$$

$$\delta X^m = \varepsilon X^m$$

$$\partial_m \left( \frac{\partial \mathcal{L}}{\partial \partial_m \varphi} \delta \varphi + \delta X^m \mathcal{L} \right) = 0$$

$$\partial_m \left[ \partial^m \varphi (-\varepsilon) (X^\alpha \partial_\alpha \varphi + \varphi) + \varepsilon X^m \mathcal{L} \right] = 0$$

$$\partial_m \left[ X_\nu \partial^m \varphi \partial^\nu \varphi + \varphi \partial^m \varphi - X^m \mathcal{L} \right] = 0 \quad \text{or}$$

Explicitly or resultados. E.g.:

e descreva melhor o porquê

$$\mathcal{D}^\mu = \partial^\mu \phi X^\nu \partial_\nu \phi + \phi \partial^\mu \phi - X^\mu \mathcal{L}$$



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④ a)  $X'^{\mu} = X^{\mu} + \delta W^{\mu}_{\nu} X^{\nu}$

$$\begin{aligned} X'^{\mu} X'_{\mu} &= (X^{\mu} + \delta W^{\mu}_{\alpha} X^{\alpha}) (X_{\mu} + \delta W_{\mu\beta} X^{\beta}) \\ &= X^{\mu} X_{\mu} + X^{\mu} \delta W_{\mu\beta} X^{\beta} + \\ &\quad + X_{\mu} \delta W^{\mu}_{\alpha} X^{\alpha} + O(\delta W^2) \end{aligned}$$

$$\delta W_{\mu\beta} X^{\mu} X^{\beta} = -\delta W_{\mu\alpha} X^{\mu} X^{\alpha}$$

$$\delta W_{\alpha\delta} X^{\alpha} X^{\delta} = -\delta W_{\delta\alpha} X^{\delta} X^{\alpha}$$

$$\delta W_{\alpha\delta} = -\delta W_{\delta\alpha}$$

OK

④ b) rotação em torno do eixo z:

$$\Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

infinitesimal:

$$\Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & \theta & 0 \\ 0 & -\theta & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \mathbb{I} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \theta & 0 \\ 0 & -\theta & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$



remarque on vérifie les 3 axes :

$$\delta W_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \theta_z & -\theta_y \\ 0 & -\theta_z & 0 & \theta_x \\ 0 & \theta_y & -\theta_x & 0 \end{pmatrix}$$

④c)

$$L = \frac{1}{2} \partial_\mu \varphi \partial^\mu \varphi - \frac{m^2}{2} \varphi^2$$

$$\varphi'(x^\mu + \delta W^{\mu\nu} x_\nu) = \varphi(x^\mu)$$

$$\delta \varphi = \varphi'(x^\mu) - \varphi(x^\mu)$$

$$= \varphi(x^\mu - \delta W^{\mu\nu} x_\nu) - \varphi(x^\mu)$$

$$= -\delta W^{\mu\nu} x_\nu \partial_\mu \varphi$$

$$\delta x^\mu = \delta W^{\mu\nu} x_\nu$$

$$\partial_\mu \left( \frac{\partial L}{\partial \partial_\mu \varphi} \delta \varphi + \delta x^\mu L \right) = 0$$



$$\partial_\mu [\partial^\mu \phi (-\delta W_{\alpha\nu} x^\nu \partial_\alpha \phi) + \delta W_{\alpha\nu} x^\nu \mathcal{L}] = 0$$

$$\partial_\mu [-\delta W_{\alpha\nu} (x^\nu \partial^\mu \phi \partial_\alpha \phi - g^{\alpha\mu} x^\nu \mathcal{L})] = 0$$

$$\partial_\mu [x^\nu \partial^\mu \phi \partial_\alpha \phi - g^{\alpha\mu} x^\nu \mathcal{L}] = 0$$

inconeto -...

$W_{\alpha\nu}$  não é arbitrário, portanto não é possível apenas removê-lo. Como exemplo:  $w_{\alpha\beta} A^{\alpha\beta} = 0$  não implica que  $A^{\alpha\beta} = 0$ , pois  $A$  pode ser um tensor simétrico, e que  $w_{\alpha\beta} A^{\alpha\beta} = 0$  implica é que a parte anti-simétrica de  $A$  é zero:

$$w_{\alpha\beta} A^{\alpha\beta} = \frac{1}{2} w_{\alpha\beta} (A^{\alpha\beta} - A^{\beta\alpha}) = 0 \Rightarrow A^{\alpha\beta} - A^{\beta\alpha} = 0$$

$$\text{neste caso: } \partial_\mu [w_{\alpha\nu} (x^\nu \partial^\mu \phi \partial_\alpha \phi - x^\nu g^{\alpha\mu} \mathcal{L})] = 0$$

$$w_{\alpha\nu} \partial_\mu [x^\nu T^{\mu\alpha}] = 0$$

$$\frac{w_{\alpha\nu}}{2} \partial_\mu [x^\nu T^{\mu\alpha} - x^\alpha T^{\mu\nu}] = 0$$

$$\Rightarrow \partial_\mu (x^\nu T^{\mu\alpha} - x^\alpha T^{\mu\nu}) = 0$$

$$M^{\mu\nu\alpha}$$



⑤ 1.6/2

Transformations infinitesimales:

$$X'^{\alpha} = X^{\alpha} + \delta W^{\alpha}_{\beta} X^{\beta}$$

$$A'^m(X') = A^m(X) + \delta W^m_{\nu} A^{\nu}(X)$$

$$\delta A_m = A'_m(X) - A_m(X)$$

$$= A_m(X^{\alpha} - \delta W^{\alpha}_{\beta} X^{\beta}) + \delta W^m_{\nu} A^{\nu}(X^{\alpha} - \delta W^{\alpha}_{\beta} X^{\beta}) - A_m(X^{\alpha})$$

$$= A_m(X) - \delta W^{\alpha}_{\beta} X^{\beta} \partial_{\alpha} A_m(X) + \delta W^m_{\nu} A^{\nu}(X) + \mathcal{O}(\delta W^2) - A_m(X)$$

$$= \delta W^m_{\nu} A^{\nu}(X) - \delta W^{\alpha}_{\beta} X^{\beta} \partial_{\alpha} A_m(X)$$

or

$$\delta X^m = \delta W^m_{\nu} X^{\nu}$$

$$\partial_m \left( \frac{\partial \mathcal{L}}{\partial \partial_m A_{\nu}} \delta A_{\nu} + \delta X^m \mathcal{L} \right) = 0$$



$$\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu}$$

$$\partial_\mu \left[ -F^{\mu\nu} (\delta W_{\nu\alpha} A^\alpha - \delta W^\alpha_\rho X^\rho \partial_\alpha A_\nu) + \delta W^\mu_\rho X^\rho \mathcal{L} \right] = 0$$

$$\partial_\mu \left[ \delta W_{\alpha\beta} \left( -F^{\mu\nu} \partial_\nu^\alpha \delta^\beta_\gamma A^\gamma + F^{\mu\nu} X^\beta \partial^\alpha A_\nu + g^{\alpha\mu} \delta^\beta_\rho X^\rho \mathcal{L} \right) \right] = 0$$

$$\partial_\mu \left[ X^\beta (F^{\mu\nu} \partial^\alpha A_\nu + g^{\alpha\mu} \mathcal{L}) - F^{\mu\alpha} A^\beta \right] = 0$$

mesmo conentário anterior:

$$\partial_\mu \left[ W_{\alpha\beta} (F^{\mu\nu} X^\beta \partial^\alpha A_\nu - F^{\mu\alpha} A^\beta + g^{\alpha\mu} X^\beta \mathcal{L}) \right] = 0$$

$$W_{\alpha\beta} \partial_\mu \left[ X^\beta (F^{\mu\nu} \partial^\alpha A_\nu + g^{\alpha\mu} \mathcal{L}) - F^{\mu\alpha} A^\beta \right] = 0$$

$$W_{\alpha\beta} \partial_\mu \left[ -X^\beta T^{\mu\alpha} - F^{\mu\alpha} A^\beta \right] = 0$$

$$\frac{W_{\alpha\beta} \partial_\mu}{2} \left[ T^{\mu\beta} X^\alpha - T^{\mu\alpha} X^\beta + F^{\mu\beta} A^\alpha - F^{\mu\alpha} A^\beta \right] = 0$$

$$\Rightarrow \partial_\mu M^{\mu\alpha\beta} = 0 \quad M^{\mu\alpha\beta}$$